Essays

|!|BCEEB0060000001|!| ★

Protein is an important food substance for humans. Either insufficient or excess intake of protein could cause undesirable effects to humans. Discuss the functions of proteins in humans and the undesirable effects may be caused if protein consumption are insufficient and in excess respectively. (10 marks)

##

Functions of proteins (Max. 3)

- As the components of various cellular structures (1)
- Involved in forming new cells for growth and repair (1)
- Form enzymes, antibodies and some hormones (1)
- As a source of energy when the body is short of carbohydrates and lipids (1)

Insufficient intake of protein (Max. 2)

- The body will break down proteins, especially those in muscles, to release energy, (1)
  leading to muscle loss. (1)
- Severe protein deficiency can lead to kwashiorkor (1)
  which causes stunted growth / anaemia / hair loss / the accumulation of tissue fluid in the abdomen. (1)

Excess intake of protein (Max. 2)

- Excess amino acids are broken down in the liver by deamination. (1)
- Eating too much protein over a long period of time would cause the liver to become overloaded, (1)
  leading to liver damage. (1)
- Excess protein intake would cause weight gain. (1)

Communication (Max. 3)

(10 marks)

##

Skills assessed	Marks
Understanding basic concepts	7
Communication	3

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|!|BCEEA0060000002|!| ★★ **STSE**

Some people adopt a vegetarian diet for its health benefits. Some people argue that adopting a vegetarian diet cannot obtain the right proportions of all the essential food substances that the human body needs. Discuss the benefits of a vegetarian diet and explain how vegetarians can maintain a balanced diet. (11 marks)

##

Benefits of a vegetarian diet

- A vegetarian diet is based on plant foods which are usually rich in dietary fibre. Obtaining enough dietary fibre can maintain normal peristalsis / prevent constipation. (1)
- Fresh fruit and vegetables usually have high vitamin C content. Obtaining enough vitamin C can prevent scurvy / help the healing of wounds / maintain healthy gums, teeth and skin. (1)
- A vegetarian diet usually contains not much lipid. This can help with weight control / prevent obesity. (1)

How vegetarians can maintain a balanced diet

- A well-planned vegetarian diet is a balanced diet. When planning a vegetarian diet, extra attention should be paid to ensure the diet includes sufficient amount of proteins, calcium, iron and vitamin B12. (1)
- Proteins can be obtained from eggs, nuts, legumes and soya products. (1)
- Calcium can be obtained from dairy products, tofu, calcium-fortified soya milk and green leafy vegetables. (1)
- Iron can be obtained from beans, tofu, spinach and some seeds such as pumpkin seeds and sesame seeds. (1)
- Vitamin B12 can be obtained from eggs, fortified soya milk and cereals. (1)

Communication (Max. 3)

(11 marks)

##

Skills assessed	Marks
Applying concepts	8
Communication	3

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|!|BCEEB0060000003|!| ★ **STSE**

In developing countries, an estimated 146 million children suffer from stunted growth due to severe malnutrition. Give an account of the likely effects of long-term severe malnutrition, other than on growth, on the body of a young child. (10 marks)

##

Effects of severe malnutrition (Max. 4)

- Low intakes of carbohydrates and fats lead to a decrease in glycogen and fat stores / severe weight loss / muscle wasting / weak muscle. (1)
- A lack of proteins leads to kwashiorkor. (1)
- Fewer enzymes / hormones / antibodies are produced, affecting metabolism and immunity. (1)
- Water potential of blood plasma increases, leading to the accumulation of tissue fluid in organs / abdomen. (1)

Effects of vitamin or mineral deficiencies (Max. 3)

- Deficiency of vitamin A leads to night blindness. (1)
- Deficiency of vitamin D leads to rickets / poor absorption of calcium. (1)
- Deficiency of vitamin C leads to scurvy. (1)
- Deficiency of iron leads to anaemia. (1)
- Deficiency of calcium leads to rickets / tooth decay. (1)

Communication (Max. 3)

(10 marks)

##

Skills assessed	Marks
Applying concepts	7
Communication	3

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|!|BCEEC0060000004|!|

Discuss the importance of vitamins and minerals to our bodies. (10 marks)

##

Importance of vitamins (Max. 4)

- Vitamin A is important for the formation of visual purple in the retina. (1)
  It can keep the cornea of the eyes, the skin, and the lining of the alimentary canal and the respiratory system healthy. (1)
- A deficiency of vitamin A may cause night blindness / thickening of the cornea / dry skin and an increased risk of infection of the lining of the respiratory system. (1)
- Vitamin C is important for the formation of connective tissues. (1)
  It also helps wound healing and maintains healthy gums, teeth and skin. (1)
- A deficiency of vitamin C may cause scurvy. (1)
- Vitamin D helps the absorption of calcium and phosphate ions from the intestine. (1)
- A deficiency of vitamin D may cause rickets and osteoporosis. (1)

Importance of minerals (Max. 3)

- Calcium is important for the formation of bones and teeth, (1)
  for blood clotting and muscle contraction. (1)
- A deficiency of calcium may cause rickets and osteoporosis. (1)
- Iron is needed in the formation of haemoglobin. (1)
- A deficiency of iron may cause anaemia. (1)

Communication (Max. 3)

(10 marks)

##

Skills assessed	Marks
Understanding basic concepts	7
Communication	3

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